

BT3041
Analysis and Interpretation of Biological Data

Term Project Guidelines

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Term Project on Large-Scale Omics Data Analysis

The aim of the term project

- To successfully demonstrate that you are able to utilize multiple concepts learnt in the course such as hypothesis testing, visual stats, dimensionality reduction, clustering, etc. while analyzing a large-scale biological datasets

Desired dataset to be used for project: Any large-scale omics datasets (transcriptome or proteome or metabolome or scRNAseq)

Typical project workflow

Dataset identification & Definition of project aims

- Choose dataset type (transcriptomics/proteomics/metabolomics/scRNA-seq)
- Download raw data and metadata from public repositories
- Define the scope of the project and define specific aims

Exploratory Analysis

- Principal Component Analysis (PCA)/UMAP visualization
- Generate plots (box plots, violin plots, heatmaps) to dissect data characteristics
- Cluster analysis (k-means, hierarchical clustering)

Hypothesis-Driven Analysis

- Differential expression (hypothesis testing)
- Correct for multiple testing using FDR methods (e.g., Benjamini-Hochberg)
- Pathway enrichment (Fisher's test, GSEA)

(or)

Classification/Regression

- Train machine learning classifiers (e.g., SVM or Random Forest) to predict sample categories
- Use regression models to identify associations between features and metadata variables (e.g., age, disease stage)

Validation & Interpretation

- Compare results with published studies to ensure biological relevance

Project reporting format

Project Presentation & Report contents

1. Introduction & Research Question(s)

- a. Background
- b. Dataset Description
- c. Research Question(s)

Examples:

- Are there significant differences in gene expression between different experimental conditions/disease states?
- Can we identify distinct clusters of samples/cells based on their omics profiles?
- What are the key features (genes, proteins, etc.) that contribute most to the variation in the dataset?
- Can we predict a specific biological outcome (e.g., drug response, disease subtype) based on the omics data?

2. Data Acquisition & Preprocessing

- a. Data Source
- b. Data Cleaning & Preprocessing

3. Methods and Codes

- Document methodology including the use of specific parameter values
- Attach codes as annexe

Project reporting format

4. Results & Interpretations

Examples:

- Explain the patterns and structures revealed in the reduced-dimensional space. Relate these findings back to the original data and your research questions.
- Describe the characteristics of the identified clusters. Relate these clusters back to the original data and your research questions. Investigate if the clusters correspond to known biological groups or conditions (if metadata is available).

5. Conclusion & Discussion

- a. Summary of Findings: Briefly summarize the key findings of your analysis in relation to your research questions.
- b. Biological Interpretation: Discuss the biological significance of your findings. What do your results suggest about the system under study?
- c. Limitations: Acknowledge any limitations of your analysis, including the dataset itself, the methods used, and any assumptions made.
- d. Future Directions: Suggest potential future research directions based on your findings and the limitations identified.

Where to look for data

Transcriptome dataset sources

1. The Cancer Genome Atlas (TCGA): (<https://www.cancer.gov/tcga>)
2. GTEx Portal (Genotype-Tissue Expression): (<https://gtexportal.org/home/>)
3. The ENCODE (Encyclopedia of DNA Elements): (<https://www.encodeproject.org/>)
4. FANTOM (Functional Annotation of Mammalian Genomes): (<https://fantom.riken.jp/>)

Proteome dataset sources

1. The Human Protein Atlas (HPA): (<https://www.proteinatlas.org/>)
2. The Human Proteome Map (HPM): (<http://www.humanproteomemap.org/>)
3. CPTAC (Clinical Proteomic Tumor Analysis Consortium): (<https://proteomic.cancer.gov/>)

Metabolome dataset sources

1. Human Metabolome Database (HMDB): (<https://hmdb.ca/>)
2. The Serum Metabolome Database: (<https://serummetabolome.ca/>)

Single-cell RNA-seq (scRNA-seq) dataset sources

1. The Human Cell Atlas: (<https://www.humancellatlas.org/>)
2. Single Cell Portal (Broad Institute): (https://singlecell.broadinstitute.org/single_cell)
3. EBI Single Cell Expression Atlas: (<https://www.ebi.ac.uk/gxa/sc/home>)
4. Deeply Integrated human Single-Cell Omics data: (<https://disco.bii.a-star.edu.sg/>)

Project timelines and milestones

Project presentation:

Milestone to be completed:

- Introduction & Research Question(s)
- Data Acquisition & Preprocessing
- Methods and Codes & Results & interpretation (at least 40%)

Target Date:

- 4 and 5 May 2026.

Duration:

- 7 mins of Presentation + 3 mins of Q&A.

Project report submission:

Milestone to be completed:

- All

Format:

- Not more than 8 pages including figures, tables and references
- Arial, font size 12, line spacing 1.
- References in APA format

Target Date:

- 19 May 2026.